

Suspected Acute Stroke Protocol — Emergency Department

Sunrise Medical Center

Document Title: Suspected Acute Stroke Protocol — Emergency Department

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Distribution: All Sunrise Medical Center Staff

1. Purpose

This protocol establishes the standardized process for the identification, triage, imaging, and treatment of patients presenting to the Sunrise Medical Center Emergency Department (ED) with signs and symptoms of an acute stroke. Rapid, coordinated action between triage nursing, emergency physicians, radiology, laboratory, and neurology is essential because eligibility for thrombolytic and endovascular therapy is time-critical.

The protocol applies to all patients arriving by ambulance, walk-in, or inter-facility transfer with a chief complaint or clinical presentation consistent with stroke, including sudden weakness, facial droop, slurred speech, vision changes, severe headache with neurological deficit, or altered consciousness of unclear origin.

2. Scope

This protocol applies to all clinical staff working within the Emergency Department, the Radiology Department, the Clinical Laboratory, and the on-call Neurology and Neurointerventional teams. Compliance is mandatory for all shifts, including nights, weekends, and holidays when specialist coverage is provided remotely via the Sunrise Telestroke Network.

3. Definitions

- **Code Stroke:** The internal activation code used to mobilize the multidisciplinary stroke response team the moment a suspected stroke patient is identified.
- **FAST Assessment:** Face drooping, Arm weakness, Speech difficulty, Time to call for help — the standardized rapid bedside screening tool used by triage staff and EMS.
- **Door-to-Needle Time:** The elapsed time between a patient's arrival at the ED and the administration of intravenous tissue plasminogen activator (tPA), target under 60 minutes.
- **Door-to-Puncture Time:** The elapsed time between arrival and groin puncture for mechanical thrombectomy candidates, target under 90 minutes.
- **NIHSS:** National Institutes of Health Stroke Scale, used to quantify neurological deficit severity.

4. Immediate Triage Actions (0–5 minutes)

Any triage nurse who identifies FAST-positive findings must immediately activate a Code Stroke overhead page and notify the charge nurse and attending emergency physician verbally, without waiting for computer documentation to be completed first. The patient is to be placed directly into a monitored resuscitation bay.

- 1 Obtain last known well time from patient, family, or EMS — this single data point determines treatment eligibility windows and must be documented to the minute.
- 2 Obtain a full set of vital signs including finger-stick glucose (hypoglycemia is a common stroke mimic and must be ruled out immediately).
- 3 Establish two large-bore IV lines and draw baseline labs: CBC, basic metabolic panel, coagulation panel (PT/INR, PTT), and type and screen.
- 4 Perform and document an NIHSS score before the patient leaves the resuscitation bay for imaging.
- 5 Page the on-call Neurology/Telestroke provider and the CT technologist simultaneously; imaging must not wait for the neurologist to arrive physically.

5. Imaging and Diagnostic Workflow

Non-contrast CT of the head is the first-line imaging study and must be initiated within 25 minutes of ED arrival for all Code Stroke activations. CT Angiography of the head and neck is obtained concurrently for patients who may be candidates for mechanical thrombectomy, particularly those with large-vessel occlusion symptoms such as gaze deviation, severe hemiparesis, or aphasia.

The radiologist on call must read the non-contrast CT within 15 minutes of image acquisition and communicate results directly to the treating emergency physician and neurologist by phone, not solely through the PACS messaging system, given the time sensitivity.

6. Thrombolytic (tPA) Administration Criteria

Intravenous alteplase may be administered to eligible patients presenting within 4.5 hours of last known well time, provided no exclusion criteria are present. The Emergency Physician and Neurologist/Telestroke physician must jointly review the exclusion checklist (Appendix A of the full Neurology Stroke Manual, SMC-NEU-002) prior to administration.

Blood pressure must be controlled below 185/110 mmHg prior to tPA administration. If elevated, use the approved antihypertensive titration order set rather than ad hoc dosing.

- Absolute contraindications include active internal bleeding, recent intracranial hemorrhage, and platelet count below 100,000.
- Weight-based dosing: 0.9 mg/kg, maximum 90 mg, with 10% given as bolus and the remainder infused over 60 minutes.
- Patient must be transferred to the ICU or dedicated Stroke Unit for the first 24 hours of neurological monitoring following administration.

7. Mechanical Thrombectomy Pathway

Patients with confirmed large-vessel occlusion on CT Angiography who present within 24 hours of last known well time (per extended-window imaging criteria) should be evaluated for transfer to the affiliated Neurointerventional Center at Riverbend Regional, as Sunrise Medical Center does not currently perform mechanical thrombectomy on site.

The transfer coordinator must be notified simultaneously with the neurology consult to avoid delay; transport arrangements should never wait for tPA infusion to complete when thrombectomy is indicated.

8. Documentation Requirements

All Code Stroke events must be documented on the Stroke Core Measures worksheet, including every timestamp from EMS dispatch through disposition. This data feeds the hospital's Joint Commission Primary Stroke Center certification reporting and is audited quarterly by the Stroke Program Coordinator.

9. Roles and Responsibilities

Role	Responsibility
Triage RN	FAST screening, Code Stroke activation, vitals and glucose
Charge Nurse	Bay assignment, staffing coordination, family communication liaison
Emergency Physician	Clinical decision-making, tPA order, exclusion criteria review
CT Technologist	Immediate imaging priority override for Code Stroke patients
Neurology/Telestroke	Remote or bedside NIHSS confirmation, treatment recommendation
Pharmacy	Rapid preparation and verification of weight-based alteplase dosing

10. Quality Assurance

The Stroke Program Coordinator reviews every Code Stroke case within five business days. Cases exceeding the 60-minute door-to-needle target are reviewed at the monthly Stroke Committee meeting to identify systemic delays and corrective actions.

11. Related Documents

SMC-NEU-002 Neurology Stroke Treatment Manual; SMC-ED-015 Code Stroke Activation Checklist; SMC-RAD-009 Emergency Imaging Prioritization Policy; SMC-PHARM-021 High-Alert Medication Verification Policy.

Appendix A: Frequently Asked Questions

Q: What if the exact last known well time cannot be determined? A: Use the time the patient was last observed to be at their neurological baseline. If truly unknown, treat the patient as outside the standard window and escalate to Neurology for individualized extended-window imaging evaluation rather than assuming ineligibility.

Q: Can a Code Stroke be activated for a patient already admitted to an inpatient unit? A: Yes. Inpatient Code Stroke follows the same activation extension (dial 5555, state 'Code Stroke' and the unit/room) and the patient is transported to CT emergently with a nurse escort; the same door-to-needle clock applies from the

time of symptom discovery.

Q: What happens if the patient or family refuses transfer for thrombectomy? A: Document the informed refusal in detail, including that the risks of not pursuing thrombectomy were explained, and notify the attending physician and Neurology immediately so alternative management can be planned.

Q: Who can activate a Code Stroke? A: Any staff member — nursing, technician, or physician — who identifies FAST-positive findings or a new focal neurological deficit may activate a Code Stroke without waiting for physician approval.

Q: Is a stat head CT required even if the patient's symptoms have already resolved? A: Yes. Transient ischemic attacks still require urgent imaging and neurology evaluation because the short-term risk of a full stroke afterward is significant.

Appendix B: Common Scenarios

Scenario — Wake-up stroke: A patient wakes up with new left-sided weakness and the last known well time is the previous night. This patient may still be eligible for advanced imaging-based selection for thrombolysis or thrombectomy; do not automatically exclude wake-up strokes from the Code Stroke pathway.

Scenario — Stroke mimic: A patient presents with FAST-positive findings but is found to be profoundly hypoglycemic. Glucose correction should occur immediately alongside, not instead of, the standard Code Stroke workup, since mimics are only confirmed after imaging and clinical reassessment.

Scenario — Inter-facility transfer: A rural clinic calls to transfer a suspected stroke patient. The ED charge nurse should activate a pre-arrival Code Stroke so imaging and neurology are ready the moment the patient arrives, rather than waiting until the patient is physically in the department.

Revision History

Version	Date	Description
1.0	Mar 3, 2023	Initial protocol release aligned with AHA/ASA guidelines
1.1	Nov 12, 2023	Added extended-window thrombectomy criteria
2.0	Feb 10, 2026	Updated door-to-needle target and Telestroke workflow